

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration in opposite <u>direction</u> to displacement or acceleration (directed) towards equilibrium position	B1
3(b)	$v = \omega(x_0^2 - x^2)^{1/2}$ and $\omega = 2\pi f$ or $v_0 = x_0 \omega$ and $\omega = 2\pi f$	C1
	substitution of any correct point from graph, e.g. for $x = 0$: $0.25 = 2\pi f \times 8.8 \times 10^{-2}$	C1
	$f = 0.45 \text{ Hz}$	A1
3(c)	$1 / 0.45 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 1.2 \text{ m}$	A1
3(d)	ellipse about the origin with same intercepts on x-axis	B1
	ellipse about the origin crossing v-axis inside original loop	B1